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Petrology and mineral chemistry of 67667, a unique feldspathic lherzolite

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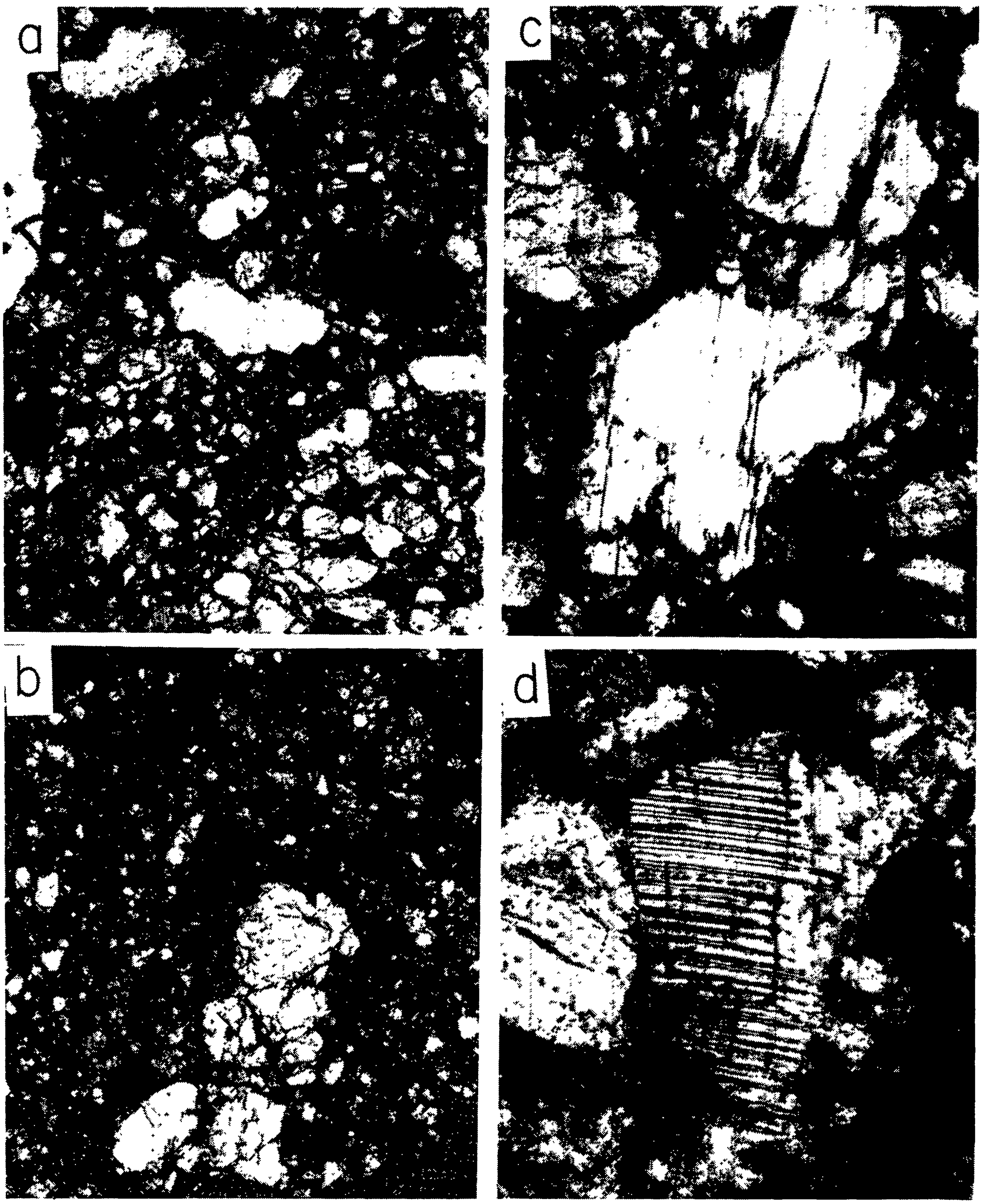
Abstract—Polished thin section 1γ of feldspathic lherzolite 67667 contains 50% olivine (Fo_{68-73}), 23% plagioclase (mainly An_{89-93}, but zoned to An_{70}), 21% low-Ca pyroxene ($En_{76}Fs_{20}Wo_4$–$En_{71}Fs_{27}Wo_2$), 5% high-Ca pyroxene (near $En_{47}Fs_{12}Wo_{41}$), 1% ilmenite (*mg* 0.15), trace spinel (high in chromite; 2.3 wt.% TiO_2). Minor elements in silicates match generally with those for silicates in highland rocks, except for the two-fold high Sr in 67667 plagioclase. There is no match with silicates from present-day meteorites. Thermometers suggest over 1000°C for pyroxene exchange and ~800°C for ilmenite-pyroxene and olivine-pyroxene exchanges. These high temperatures and lack of evidence suggesting metamorphic textures tentatively suggest crystallization at shallow depth in the lunar crust.

INTRODUCTION

Samples returned from the surface of the moon may have been excavated from some tens of kilometers depth by impacting bodies. These events should also have brecciated, melted and metamorphosed samples to varying extents so that present textures and chemistry may not bear much resemblance to their precursors. Unusual chemical or mineralogical features of samples may be the only clues to rare or seldom sampled petrogenetic processes on the moon. Below we describe a unique feldspathic lherzolite sample and speculate on its possible origin.

Rake sample 67667, weighing 7.9 gm, was initially described by Steele and Smith (1973) and recognized as unusually mafic-rich. Although similarities to some mare samples were noted, minor element contents of plagioclase were unlike those of described mare plagioclase, and mafics showed very restricted compositional variations. Warren and Wasson (1978) suggested that 67667 was probably pristine based on inferred coarse grain size and low Ni/Co ratios of 4 metal blebs. Warren and Wasson (1979) confirmed that 67667 was pristine based on siderophile data and an extremely fractionated pattern of incompatible elements relative to KREEP. Analyses of minor and trace elements of plagioclase and olivine were briefly reported in Hansen *et al.* (1979a,b), Hansen *et al.* (1980) and Steele and Smith (1979, 1980) as part of a systematic study of mineral chemistry of pristine rocks. A bulk analysis of fragment 67667,3 was given by Warren and Wasson (1979), and a mode and major-element mineral compositions were given for thin section 67667,6. The norm (Warren and Wasson, 1979) (58% olivine, 15% opx, 5% "diopside", 21% feldspar) shows that 67667 is the most ultramafic of the Apollo 16 rocks (Norman and Ryder, 1979). Although most mineral properties allow 67667 to be classified with the Mg-rich plutonic trend (Warner *et al.*, 1976; Warren and Wasson, 1979), the Sr content of the plagioclase is about twice



that for the single trend encompassing plagioclase from all other measured rock types (Steele and Smith, 1979). Furthermore, the flat rare-earth pattern distinguishes 67667 from all other pristine rocks except norites 73255,27,65 and 77035,130 (Norman and Ryder, 1979). High-Ca pyroxene is rare in lunar rocks, and only three gabbros (61224,6; 67915c; 76225c) have a high content of Ca-rich pyroxene and a low abundance of plagioclase like 67667 (Marvin and Warren, 1980).

PETROGRAPHY AND MINERAL CHEMISTRY

Because optical identification of shocked silicates is difficult, a point count of 67667,1γ was made with the aid of a solid-state detector on an electron microprobe. The resulting mode (olivine 50%, plagioclase 23%, low-Ca pyroxene 21%, high-Ca pyroxene 5%, ilmenite 1%) agrees well with the norm for 67667,3 (Warren and Wasson, 1979).

Approximately half of the 5mm^2 area of 67667,1γ is shown in photomicrographs (a) and (b) of Fig. 1. Although most grains of the cataclastic texture are small and fractured with irregular boundaries, a few larger grains in clasts appear to have survived sufficiently to allow approximate reconstruction of the original texture. The polymineralic clast in (a) contains both olivine and pyroxene grains with a tendency to elongated-hexagonal outline up to 0.3 mm long and 0.2 mm wide. There is a weak tendency towards parallelism of the elongated-hexagonal grains, and the texture may result from partial fracturing of an original cumulus texture. A monomineralic area, 0.9 × 0.5 mm, contains several grains of low-Ca pyroxene with near-parallel extinction (b) suggestive of disruption of a single crystal. Warren and Wasson (1978) reported several regions 2 mm across and nearly monomineralic in 67667,6 which has an area about 10 times larger than 67667,1γ. A plagioclase clast in the center of (a), and enlarged in (c), is the remnant of a grain at least 0.3 mm wide. Although it is impossible to give a

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Fig. 1. Photomicrographs of 67667,1γ. (a,b) 1.5 × 1.0 mm with partly-crossed polars; (c) 0.4 × 0.25 mm, (d) 0.15 × 0.1 mm, both with crossed polars. Transmitted light. (a) The clast at the lower right shows euhedral to subhedral pyroxene and olivine crystals in a complex texture which may result from settling of grains into a cumulus texture followed by partial fracturing along cleavages and partings. Some small crystals between larger crystals have wedge shapes. The largest ilmenite crystal is one-third up the left-hand side. (b) A clast fills most of the area. Strong fracturing has obscured the original texture, but several grains have extinction positions within about 10° and irregular transitional boundaries suggestive of derivation from a single crystal. (c) One-third of the area is covered by plagioclase crystals which are probably fractured relics of a single crystal. Rims zoned to An_{70} are shown by r. Twin lamellae are offset in the largest crystal, and multiple tapered twin lamellae occur in the crystal at top center. Most other crystals are pyroxenes, but a small twinned plagioclase crystal occurs near the base of the left-hand margin. (d) Low-Ca pyroxene crystal with multiple lamellae of high-Ca pyroxene. The boundaries are wavy.

Table 1. Electron microprobe analyses of 67667,1γ.

	1	2	3	4	5	6	7	8
P_2O_5	*0.009*	nd	*0.015*	*0.013*	nd	nd	nd	nd
SiO_2	37.6	37.5	53.8	52.2	45.4	51.4	0.5	0.2
TiO_2	*0.049*	nd	*0.56*	*1.42*	nd	nd	52.6	2.3
Al_2O_3	*0.017*	nd	*1.50*	*2.16*	35.3	29.8	0.3	15.8
Cr_2O_3	*0.013*	nd	*0.67*	*0.87*	nd	nd	nd	46.8
FeO	25.2	28.4	14.2	7.21	*0.110*	0.4	42.0	30.0
MnO	*0.26*	0.3	*0.21*	*0.15*	nd	nd	0.4	nd
MgO	37.2	34.6	27.4	16.8	*0.070*	nd	4.3	4.9
NiO	*<0.015*	nd	*<0.015*	*<0.015*	nd	nd	nd	nd
CaO	*0.054*	nd	1.8	20.4	18.5	14.7	nd	nd
Na_2O	*<0.002*	nd	*0.013*	*0.078*	0.61	2.2	nd	nd
K_2O	nd	nd	nd	nd	*0.078*	0.4	nd	nd
Total	100.402	100.8	100.168	101.301	100.068	98.9	99.6	100.0

1 typical olivine, Fo_{72}; 2 rare Fe-rich olivine, Fo_{68}; 3 low-Ca pyroxene, average of 11 analyses, $En_{74}Wo_4$; 4 high-Ca pyroxene, average of 5 analyses, $En_{47}Wo_{41}$; 5 typical plagioclase, average of 11 analyses, An_{92}; 6 sodium-rich plagioclase at rim of grain, An_{72}; 7 ilmenite, average of 14 analyses; 8 spinel.

nd not detected in energy-dispersive system. High-precision analyses with spectrometers shown by italics.

definitive statement on the original texture, 67667 appears to have been a fine-grained rock with most grains on the order of a millimeter in size. There is no evidence of metamorphic recrystallization after cataclasis, as was inferred for lunar specimens with granoblastic or poikiloblastic textures (e.g., Bickel and Warner, 1978).

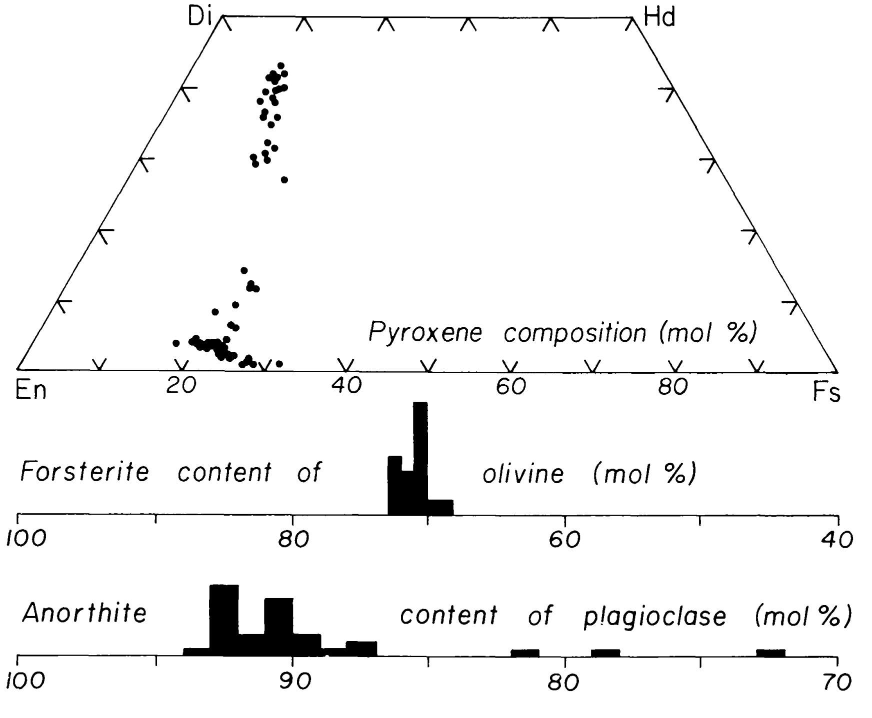

Fig. 2. Major-element chemistry of silicate minerals in 67667,1γ.

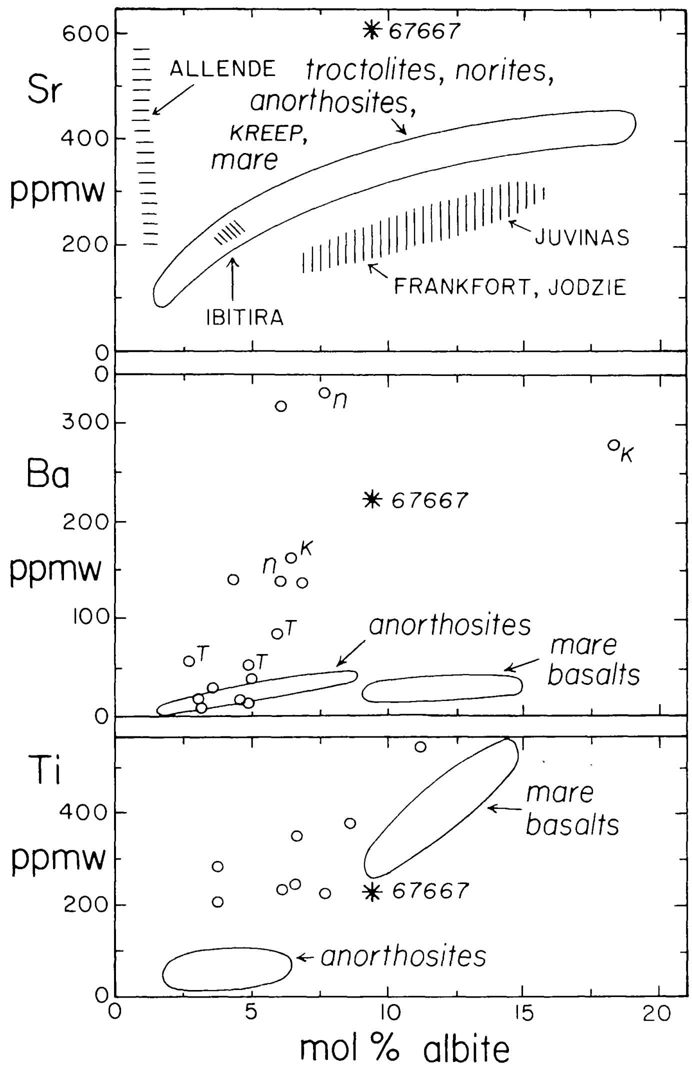

Fig. 3. Comparison of minor-element chemistry of plagioclase from 67667 and selected lunar and meteoritic plagioclases. Details are given in Steele *et al.* (1980).

Plagioclase grains in section 1γ show several signs of shock: fractures, microfaults that offset twin lamellae, multiple tapered twins, and undulatory extinction. Only one grain shows any sign of maskelynitization, in contrast to section 6 for which most grains were described as maskelynitized (Warren and Wasson, 1978). Most of the large grain in (c) has a restricted range of 2% An around a median of 91% An, which overlaps with the range 90.3–93.0% for section 6. At two boundaries, normal zoning occurs to 72% An (Table 1, anal. 6) over 20 μm. It is inferred that zoning continued around the crystal, and that the other zoned boundaries were fractured away (see legend to Fig. 1c). All other plagioclase grains are small and have irregular edges with no evidence of zoning. Most analyses lie between 87 to 94% An (Fig. 2). Minor-element analyses were made as in Hansen *et al.* (1979a). The ranges of Fe (0.029–0.262 wt.%), Mg (0.034–0.17 wt.%) and mg = Mg/(Mg + Fe) ratio (0.41–0.68; mean 0.54) are much larger than the scatter from counting statistics for the mean value of 11 analyses in Table 1, col. 5. Fluorescence might cause a systematic error for Fe in spite of avoidance of Fe-rich minerals, but could not affect Mg because of the insignificant fluorescent yield for this light element.

Fig. 3 summarizes the key features of ion microprobe analyses of plagioclase in 67667 (Steele and Smith, 1979, 1980). The Sr analysis is about twice that found for plagioclase in the single trend for anorthosites, mare basalts, KREEP-rich rocks, norites and troctolites: new analyses confirmed the original analyses. The Ba content is higher than for plagioclases from anorthosites and mare basalts, but is within the wide spread for plagioclases from norites and KREEP-rich rocks. The Ti content lies in the middle of the range, and the Li content (not shown) is relatively low.

A few grains of low-Ca pyroxene display fine exsolution lamellae about 1 μm across (Fig. 1d). Major-element analyses with a narrow electron beam and an EDS system gave distinct composition regions for the low-Ca and high-Ca pyroxenes (Fig. 2). The spread can be attributed in part to beam overlap onto host and lamellae. For low-Ca pyroxene, most of the analyses lie in a narrow band from $En_{76}Fs_{20}Wo_4$ to $En_{71}Fs_{27}Wo_2$, and analyses with >5% Wo are attributed to overlap with lamellae of high-Ca pyroxene. For high-Ca pyroxene, most analyses lie in a band pointing towards low-Ca pyroxene without any indication of zoning to Fe-rich compositions. The center of the range fits fairly well with that found for section 6, but the analyses for section 1γ extend to higher Wo content. Although genuine chemical zoning may be responsible, overlap onto sub-microscopic lamellae of low-Ca pyroxene is likely in view of the evidence for visible lamellae of high-Ca pyroxene. Table 1, column 4, gives the average of the 5 analyses with highest Wo content; other analyses may represent two phases.

Special spectrometer analyses for minor elements in pyroxenes are given in Table 1, columns 3 and 4. A survey of pyroxene compositions in the literature shows that the titanium and aluminum contents of the low-Ca pyroxene and the titanium and chromium contents of high-Ca pyroxene in 67667 are higher than in most pristine highland rocks, but not exceptionally so. Defect-free atomic formulae can be calculated within experimental error without changing Cr from assumed trivalency to divalency. One reviewer (C. Herzberg) noted that for the structural formula for the low-Ca pyroxene composition given in Table 1, column 3, some Al can be assigned to the Ml site. Other lunar samples for which this is possible include troctolite 76535, anorthosite 62275 and dunite 72415 but not for rocks resulting from impact melting and rapid cooling (Herzberg, 1979).

Olivines range only from 68 to 73% Fo with a maximum at 70.5. Special minor-element analyses (Table 1, col. 1) are unexceptional with respect to those for olivines from other highlands rocks (Hansen *et al.*, 1980). The Cr content in 67667 olivine is lower than for olivines in other Mg-rich rocks in spite of the presence of Cr, Al-rich spinel, but is higher than for olivines in ferroan anorthosites.

In general, the major-element chemistry of the silicates in 67667 fits better with noritic and related specimens listed in Smith and Steele (1976, Fig. 9) than with other specimens. The major difference between 67667 and norites 72255 and 78235 is a smaller gap between pyroxene compositions in 67667. The most Wo-rich analyses on Fig. 2 represent the Ca-rich pyroxene in 67667 without overlap onto Ca-poor pyroxene.

Several ilmenite grains up to 0.1 mm were found, and tiny spinel grains are contiguous to many of them. The ilmenite (Table 1, col. 7) is nearly homogeneous, with *mg* 0.14–0.16. The spinel (col. 8) is rich in the chromite end-member, and contains only 2.3% TiO_2. Its composition falls in the range for highland samples in Figs. 12 and 14 of Smith and Steele (1976) with a fairly close fit to spinel from cataclastic anorthosite 60215. Highly-reflecting grains, 1–5 μm across, are scattered throughout the thin section, but were not analyzed in detail. Presumably they would prove to be Fe metal with high Co/Ni, as found by Warren and Wasson (1978) and Ryder *et al.* (1980): the ranges of 0.8–4.0 Co and 3–7 wt.% Ni overlap with ranges for noritic and anorthositic metal in Figs. 20 and 21 of Smith and Steele (1976).

THERMOMETRY

Application of the clinopyroxene Ca/Mg/Fe thermometer is uncertain because of the range of compositions in Fig. 2, and the problem of overlap onto fine lamellae. The mean composition $En_{47}F_{12}Wo_{41}$ from Table 1 derived using the most Ca-rich analyses yields a temperature of 1060°C for the Wood and Banno (1973) formula, and 1070°C for the Wells (1977) formula. Higher values would be obtained as the Wo content of the high-Ca pyroxene is reduced to the mean value of $\sim En_{51}Fs_{13}Wo_{36}$ for all 24 points in Fig. 2.

The distribution of magnesium and iron between ilmenite and pyroxene (Bishop, 1976, 1980) yields 723 to 781°C for low-Ca pyroxene $En_{74}Fs_{22}Wo_4$ and the range *mg* 0.14–0.16 for ilmenite. Using high-Ca pyroxene $En_{47}Fs_{12}Wo_{41}$ yields 820 to 880°C.

The average Ca content of olivine (0.54 wt.%) yields a temperature of slightly less than 800°C from the experimental data of Finnerty and Boyd (1978) and the reasonable assumption that the pressure at which 67667 crystallized was not more than a few kilobars.

It is not clear what are the causes of the discrepancies among the above temperatures. Although there must be experimental errors and theoretical problems of unknown nature, the main cause may be differences in the blocking temperatures for the different exchange reactions. Steele *et al.* (1977) found that the Bishop (1976, 1980) thermometer for coexisting oxide and silicate minerals in the Fiskenaesset complex gave a temperature about 100°C less than for the Wood-Banno pyroxene thermometer. Bishop (1980) obtained differences between the orthopyroxene-ilmenite and clinopyroxene-ilmenite thermometers. Usselman (1975) reported rapid equilibration between ilmenite and other phases during experimental cooling of mare basalts.

In spite of uncertainty over details, it seems safe to conclude that 67667 did not equilibrate to low temperatures (~600–800°C) usually associated with an-

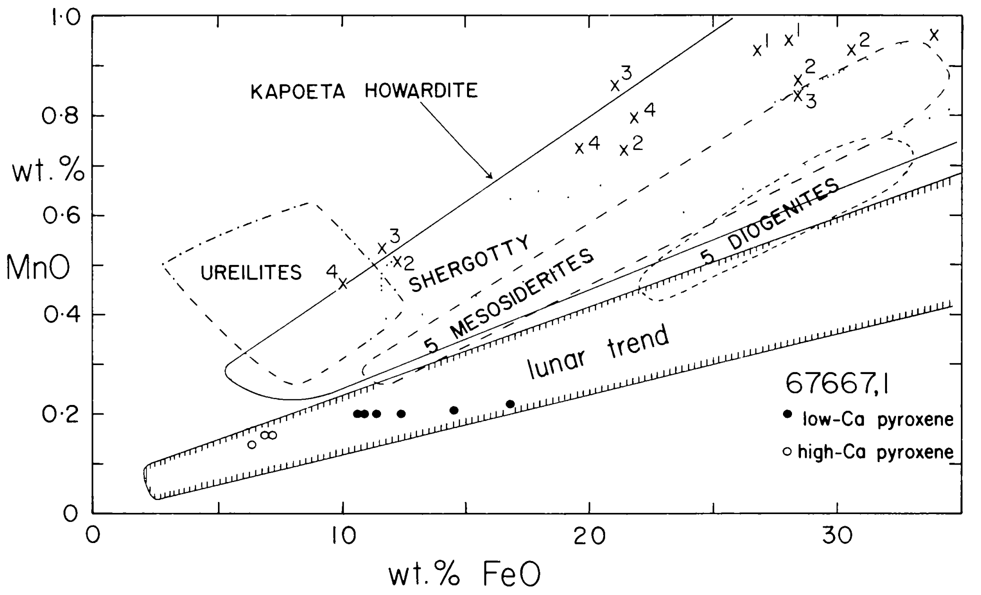

Fig. 4. Comparision of MnO and FeO contents of pyroxenes from 67667 and selected meteorites. The known range for lunar pyroxenes is outlined. Eucrites: crosses; 1 Simon and Haggerty (1980), 2 Miyamoto *et al.* (1976), 3 Takeda *et al.* (1976), 4 Takeda *et al.* (1979). Shergottite: dot; Smith and Hervig (1979). Howardite: full-line; Dymek *et al.* (1976). Mesosiderites: dash; Agosto *et al.* (1980). Ureilites: dot-dash; Takeda *et al.* (1980).

nealing for an extended period in a plutonic environment. Neither we nor Warren and Wasson (1978) recognize remnant equilibrium textural features (although present textures are cataclastic) suggestive of long term annealing. We thus infer a near-surface environment which allowed relatively rapid cooling and thus preventing metamorphic equilibration.

PROVENANCE

Because of the uniqueness of 67667, it is necessary to raise the question whether it was formed in the moon or whether it is a "meteorite". Unfortunately there is the semantic problem of defining what is meant by a meteorite because of the difficulty of drawing a boundary between the end of accretion and the beginning of meteorite bombardment. This problem has already arisen about basin-forming projectiles, for which Hertogen *et al.* (1977) proposed that the chemical composition was less moon-like for the later impacts. For clarity, two questions will be raised: (a) does 67667 have properties that can be either matched with present-day meteorites, or related to them by plausible crystal-liquid differentiation, and (b) does 67667 have properties that can be related to the major groups of lunar rocks?

Two pieces of evidence indicate that 67667 is not related to known meteorites. The Sr vs. Ab plot (Fig. 3) shows that 67667 plagioclase cannot be matched with plagioclase from the howardites and eucrites studied so far. Plagioclase from the Allende meteorite is too rich in anorthite molecule, and plagioclases from all other meteorites, including shergottites and both O- and E-chondrites, are too rich in albite molecule. Figure 4 shows that the Fe/Mn ratio of ferromagnesian minerals from 67667 lherzolite falls in the lunar trend and not in the regions occupied by known meteorites.

In contrast to the lack of correspondence with meteorites, there is a general relationship between 67667 lherzolite and several lunar rocks. The closest mineralogical relationship is with norites (preceding section) and gabbros. However, the Ti/Sm ratio of 67667 as illustrated by Norman and Ryder (1980) is unlike nearly all other pristine norites and troctolites but similar to pristine anorthosites. Other chemical parameters such as Sr-Ab in plagioclase (Fig. 3) and Sc/Sm (Norman and Ryder, 1980) do not agree well with other petrologic groups. Tentatively, we prefer to explore the possibility that 67667 is a representative of a new family of lunar rocks. Many authors have speculated that the lunar crust contains distinct plutons, and perhaps 67667 represents such an isolated body. The apparent non-equilibrium texture, the chemical zoning of the plagioclase and the high blocking temperatures for the exchange reactions, are all consistent with crystallization at a shallow level in the lunar crust. Absence of a europium anomaly in 67667 (Warren and Wasson, 1979) indicates that the parent liquid had not lost plagioclase. The ten-fold enrichment of rare earths over chondrites is comparable to that in lunar norites. Two other gabbros, 67915c and 61224,6 (Marvin and Warren, 1980), are also rich in high-Ca plagioclase. Detailed studies of mineral chemistry are needed to explore possible relationships between these gabbros, norites and 67667 lherzolite.

CONCLUSIONS

The petrography and mineral chemistry of 67667 lherzolite indicates cataclasis of a fine-grained high-temperature rock, perhaps formed as a cumulate in a high-level pluton. Except for the Sr content of plagioclase, the mineral chemistry fits with that of major rock types ascribed to the lunar crust. There is no evidence to favor a relationship between 67667 and present-day meteorites falling on the earth.

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